

Feasibility Study

Feasibility is defined as the practical extent to which a project can be performed successfully. To evaluate feasibility, a feasibility study is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software. Information such as resource availability, cost estimation for software development, benefits of the software to the organization after it is developed and cost to be incurred on its maintenance are considered during the feasibility study. The results of the feasibility study should be a report that recommends whether or not it is worth carrying on with the requirements engineering and system development process.

If a system does not support the business objectives, it has no real value to the business. While this may seem obvious, many organisations develop systems which do not contribute to their objectives either because they don't have a clear statement of these objectives, because they fail to define the business requirements for the system or because other political or organisation factors influence the system procurement.

The objective of the feasibility study is to establish the reasons for developing the software that is acceptable to users, adaptable to change and conformable to established standards. Various other objectives of feasibility study are listed below.

- To analyse whether the software will meet organizational requirements
- To determine whether the software can be implemented using the current technology and within the specified budget and schedule
- To determine whether the software can be integrated with other existing software.

The information assessment phase identifies the information that is required to answer the three questions set out above. Once the information has been identified, you should question information sources to discover the answers to these questions. Some examples of possible questions that might ask:

1. How would the organisation cope if this system was not implemented?

Answer: Without MediSense, the clinic would continue to rely on manual follow-ups. With ~70% of chronic patients (Hypertension & Diabetes) exhibiting non-adherence due to lack of symptoms or side-effect fears, patient health deteriorates silently. This eventually leads to high rates of preventable emergency hospitalizations after 1 to 2 months of missed medication.

2. What are the problems with current processes and how would a new system help alleviate these problems?

Answer: Current processes lack any systematic tracking software post-discharge, and busy doctors cannot personally follow up with every patient. MediSense alleviates this by automatically tracking patient logs and surfacing predictive dashboard alerts for high-risk patients before emergency complications arise.

3. What direct contribution will the system make to the business objectives and requirements?

Answer: It directly reduces long-term chronic disease complications (stroke, heart attack, renal damage), saves clinical time through prioritized high-risk alerts, and improves overall quality of patient care.

4. Can information be transferred to and from other organisational systems?

Answer: Yes, MediSense uses standard REST APIs and relational database architecture (Django + PostgreSQL), making it fully capable of integrating with existing hospital management systems or Electronic Health Records (EHR).

5. Does the system require technology that has not previously been used in the organisation?

Answer: No, it relies on industry-standard, web-accessible open-source technology (Django, Python, HTML/Tailwind). Clinics only need a standard web browser to access the system.

6. What must be supported by the system and what need not be supported?

Answer:

- **Must support:** Patient registration, daily dose logging, missed dose tracking, risk classification, and predictive alerts for doctors.
- **Need not support:** Inpatient bed management, hospital billing, or pharmacy inventory management.

Types of Feasibility

Various types of feasibility that are commonly considered include technical feasibility, operational feasibility, and economic feasibility.

Technical feasibility assesses the current resources (such as hardware and software) and technology, which are required to accomplish user requirements in the software within the allocated time and budget. Technical feasibility also performs the following tasks.

- Analyses the technical skills and capabilities of the software development team members
- Determines whether the relevant technology is stable and established
- Ascertains that the technology chosen for software development has a large number of users so that they can be consulted when problems arise or improvements are required.

Operational feasibility assesses the extent to which the required software performs a series of steps to solve business problems and user requirements. This feasibility is dependent on human resources (software development team) and involves visualizing whether the software will operate after it is developed and be operative once it is installed. Operational feasibility also performs the following tasks.

- Determines whether the problems anticipated in user requirements are of high priority
- Determines whether the solution suggested by the software development team is acceptable

- Analyzes whether users will adapt to a new software
- Determines whether the organization is satisfied by the alternative solutions proposed by the software development team.

Economic feasibility determines whether the required software is capable of generating financial gains for an organization. It involves the cost incurred on the software development team, estimated cost of hardware and software, cost of performing feasibility study, and so on.

- Cost incurred on software development to produce long-term gains for an organization
- Cost required to conduct full software investigation (such as requirements elicitation and requirements analysis)
- Cost of hardware, software, development team, and training.